

Rings and Subrings. Choose any 3.

1. Rings: *Give an example for each of the following.*
 - a. A ring with identity
 - b. A ring without identity
 - c. A non-commutative ring
 - d. A subring of $\mathbb{Z}_6$, $\mathbb{Q}$, and $\mathbb{C}$.
2. Prove that the set of nonnegative integers is not a subring of $\mathbb{Z}$.
3. Prove that $\mathbb{Z}[i] = \{a + bi : a, b \in \mathbb{Z}\}$ is a subring of $\mathbb{C}$.
4. Let R be a ring with subrings A and B . Prove that $A \cap B$ is also a subring.

Integral Domains and Fields. Choose any 3.

5. Integral domains and Fields: *Give an example for each of the following.*
 - a. An integral domain.
 - b. A ring which is not an integral domain.
 - c. A field.
 - d. A ring which is not a field.
6. Let R be an integral domain and suppose that $ax - bx = 0$. Prove that if $a \neq b$ then $x = 0$.
7. Let $R = \mathbb{Z}_{10}$
 - a. Prove that R is not an integral domain.
 - b. Prove that R is not a field.
8. Let $R = \mathbb{Z}_3$
 - a. Prove that R is a field.
 - b. Prove that R is an integral domain.

Ideals. Choose any 3.

9. Ideals: *Give an example for each of the following.*
 - a. A ring and an ideal in that ring.
 - b. A ring and a subring which is not an ideal.
 - c. A ring other than $\mathbb{Z}$, and a principal ideal in that ring.
 - d. A ring other than $\mathbb{Z}_{12}$, and a prime ideal in that ring.
10. Prove that for any ring R , 0_R is an ideal of R .
11. Prove that $E = \{x : x = 2n \text{ for } n \in \mathbb{Z}\}$ is an ideal in $\mathbb{Z}$.
12. Find all the ideals and maximal ideals of $\mathbb{Z}_{12}$ (use a lattice).

Homomorphisms and Isomorphisms. Choose any 3.

13. Homomorphism and Isomorphisms: *Give an example for each of the following.*

- a. A function between rings which is not 1-1.
- b. An onto function between rings.
- c. An isomorphism between rings (other than the identity function).
- d. A homomorphism between rings which is not an isomorphism.

14. Prove that the rings $\mathbb{Q}$ and $\mathbb{Z}$ are not isomorphic.

15. Prove that $\mathbb{Z}$ is not isomorphic to $\mathbb{Z}_6$.

16. Let $f: R \rightarrow S$ be a ring homomorphism.

- a. What is $f(0_R)$?
- b. Suppose that $f(a)f(b) = 1_R$ find $f(ab)$.
- c. Suppose that $\ker f = \{0, r\}$ for some nonzero $r \in R$. Is R isomorphic to S ?

Factor Rings. Choose any 3.

17. Ideals: *Give an example for each of the following*

- a. A factor ring.
- b. Two elements in the factor ring above.
- c. An abelian factor ring.
- d. A factor ring that is a field.

18. Let $F = \{x: x = 4n + 1 \text{ for } n \in \mathbb{Z}\}$, the set of odd integers.

- a. Prove that F is not an ideal in $\mathbb{Z}$.
- b. Prove that $\mathbb{Z}/F$ is not a ring.

19. Consider $\mathbb{Z}_{10}/\langle 2 \rangle$.

- a. Give two examples of distinct elements in $\mathbb{Z}_{10}/\langle 2 \rangle$.
- b. Prove that $\mathbb{Z}_{10}/\langle 2 \rangle$ is an integral domain

20. Define $f: \mathbb{Z} \rightarrow \mathbb{Z}_6$ as $f(x) = x \bmod 6$.

- a. Let $f: \mathbb{Z} \rightarrow \mathbb{Z}_6$. Find the kernel of f .
- b. Show that $\mathbb{Z}/6\mathbb{Z}$ is isomorphic to $\mathbb{Z}_6$.

21. Recall that $\langle i \rangle$ is a principal ideal in $\mathbb{Z}[i]$. Suppose that the function $f: \mathbb{Z}[i] \rightarrow \mathbb{Z}$ is an onto homomorphism with kernel, $\ker f = \langle i \rangle$. Prove that $\mathbb{Z}[i]/\langle i \rangle$ is isomorphic to $\mathbb{Z}$.